

Automation System Design: Dual-Conveyor Sorting System

1. Introduction

This report presents the design of an automated material sorting system developed in RoboDK. The system utilizes an ABB CRB 1300 robot, two independent conveyor belts, and photoelectric sensors to manage material flow efficiently.

2. System Overview

The simulation environment consists of:

- **Robotic Arm:** ABB CRB 1300 industrial collaborative robot.
- **Dual Conveyor Architecture:** Conveyor A (input/staging) and Conveyor B (secondary routing), allowing for dynamic material sorting.
- **Sensors:** Laser Sensor A and Laser Sensor B, acting as triggers for the robot's pick-and-place logic.

3. Automation Logic Flow

The control logic operates on an event-driven basis:

- **Detection:** Laser sensors continuously monitor the belt surface for incoming materials (boxes).
- **Trigger:** Once an object interrupts the laser beam (*Signal = True*), the controller pauses the belt or initiates the robot's pick sequence.
- **Execution:** The robot follows a predefined path (MoveJ/MoveL) to the detected coordinate and performs the transfer between conveyors.

4. Implementation Analysis

The simulation demonstrates successful synchronization between material arrival and robotic intervention. By leveraging sensor-based feedback, the system maintains high precision and reduces idle time, illustrating core principles of Industry 4.0 automation.

5. Conclusion

The automated sorting system is fully functional and optimized for flexible material handling. This project confirms the viability of sensor-integrated robot cells for real-world manufacturing applications.